Project handbook

Wetterstation 001

Version 19

Project manager: Matthias Brückl

Date: 12/06/26

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Document versions

Versionno.	Date	Change	Author
1	24.11.25	Start	Brückl M.
2	28.11.25	Objectives and project organization completed	Aistleitner D.
3	29.11.25	Completion till 1.5	Bergbauer K.
4	30.11.25	Completion till 1.9	Bergbauer K. Aistleitner D.
5	01.12.25	Updateing 1.4 and 1.9	Aistleitner D.
6	02.12.25	Finishing the Parts for Project Handbook Part 1 up to 1.10	Brückl M.
7	24.01.26	Optimizing project environment graphic	Bergbauer K.
8	25.01.26	Update according the feedback	Brückl M.
9	01.02.26	Update according the feedback and complete 1.13 to 2	Brückl M.
10	02.02.26	First Weekly Report	Bergbauer K.
11	16.02.26	Weekly Report + Update according the feedback	Brückl M.
12	11.03.26	Weekly Report + Update according the feedback	Aistleitner D.
13	18.03.26	Write next Weekly Report	Brückl M.
14	24.03.26	Weekly Report + Update according the feedback + new WBS	Bergbauer K.
15	14.04.26	Write next Progress Report	Aistleitner D.
16	29.04.26	Weekly Report + Update according the feedback	Brückl M.
17	13.05.26	Weekly Report	Bergbauer K.
18	27.05.26	Weekly Report	Aistleitner D.
19	03.06.26	Weekly Report	Brückl M.
20	10.06.26	Project close	Aistleitner D. & Brückl M.

Distribution list

Name	Organisational Unit	Role in project	Telephone	E-mail
Brückl Matthias	Project Manager	Backend Developer	0664 994484299	Matthias.brueckl@htl-wels.at
Berbauer Klaus	Project Team Member	Hardware and 3d Modelling	0677 641591985	Daniel.aistleitner@htl-wels.at
Aistleitner Daniel	Project Team Member	Frontend Developer	0660 90940434	Klaus.bergbauer@htl-wels.at
HTl-Wels	Project Owner		0664123456789	office@htl-wels.at
Smart-Wand	Stakeholder	Help with optimizing	0681 108468979	Paul.auer@htl-wels.at
Smart Flowerpot	Stakeholder	Help with optimizing	0677 632238199	Thomas.toferer@htl-wels.at

1 Project plans

1.1 Project Assignment

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PROJECT-ASSIGNMENT

Project start event:
- Submission of the Project shandbook

Project start date:
- 12.01.2026

project close down event in terms of content:
- Active usage of the weatherstation

Formal project close down event:
- Handing the weatherstation over to the teacher

Project close down dates:
- 10.06.2026

Project objectives:
- Functioning weather station
- Energy-independent power supply
- Web based data analyse
- Minimalist Case

Non-objectives:
- No complex data processing
- No guarantee of absolute accuracy of the
- measurement data.
- No use of cloud storage (e.g., Google Drive, AWS,
- Azure).
- No failover protection through redundant hardware
- components.
- No remote control

Main tasks (Project phases):
- Conding Frontend
- Read out the date of the sensors
- Coding a small database
- Coding APIs
- 3d modelling the Case

Project resources and costs*:

resource/type of cost	unit	Costs (€)
Raspberry Pi 2w / A	1	30€
BME280 / A	1	5€
DHT22 / A	1	8€
VEML6075 / A	1	10€
Breadboard / A	1	5€
PETG-Filament / A	3	45€

Project owner:
- Htl Wels

Project manager:
- Brückl Matthias

Project team members:
- Aistleitner Daniel
- Bergbauer Klaus

HTL Wels, (Project owner)

Matthias Brückl (Project manager)

* Possible categories of total Project budget:
Category A: up to 0,3 million Euro
Category B: up to 1 million Euro
Category C: up to 10 million Euro
Category D: more than 10 million Euro

1.2 Project Objectives (objectives, non-objectives)

Wetterstation 001	PROJECT OBJECTIVES	
Type of objective	**Project objectives**	**Adjusted project objectives as of...**
objectives: • **Main objectives** • **Additional objectives**	• Readout following sensors: • UV sensor • Temperature sensor • Air pressure sensor • Humidity sensor • Anempmeter for windspeed • Web based data analyse • Minimalist Case	•
Non-objectives	• No complex data processing • No guarantee of absolute accuracy of the measurement data. • No use of cloud storage (e.g., Google Drive, AWS, Azure). • No failover protection through redundant hardware components. • No remote control	• If there is enough time left before end of mai this objective could be transformed into an additional object.

1.3 Description of Pre- and Post Project Phase

Wetterstation 001	DESCRIPTION OF PRE- AND POST-PROJECT PHASE

- **Pre-project phase**

What triggered the project?
- An assignment for ITP in school
- The idea of the project was given to us by Brückl's mom
- She wants to know the exact temperature in her garden

Relevant documents for the project („Minutes", … ONLY documents and no content necessary)
- Weatherstation_Goals.docx
- Materiallist.docx
- Wetterstation_Pflichtenheft.docx
- Protokoll.docx

Experience from similar projects
- Groupwork SEW in second class (Aistleitner/Brückl/Bergbauer)
- Built a react Website (Brückl)

- **Post-project phase**

What will happen after the project has ended? (follow-up activities, further projects, …)?
- Project will be presented on the "Open House Day" at School

1.4 Project Environment Analysis

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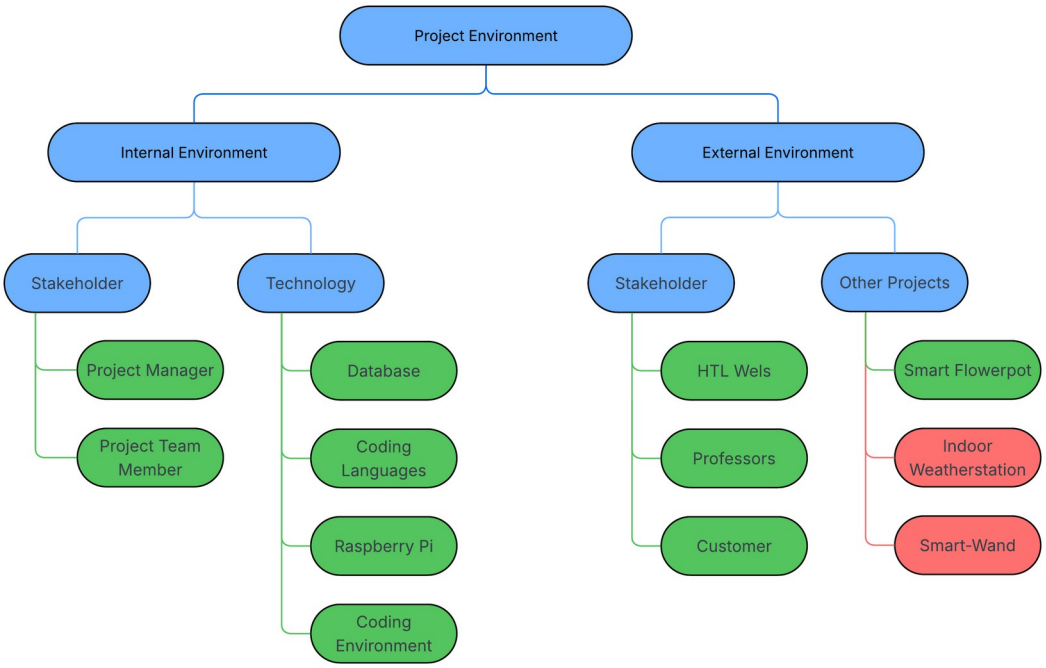
PROJECT ENVIRONMENT GRAPHIC



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PROJECT ENVIRONMENT TABLE

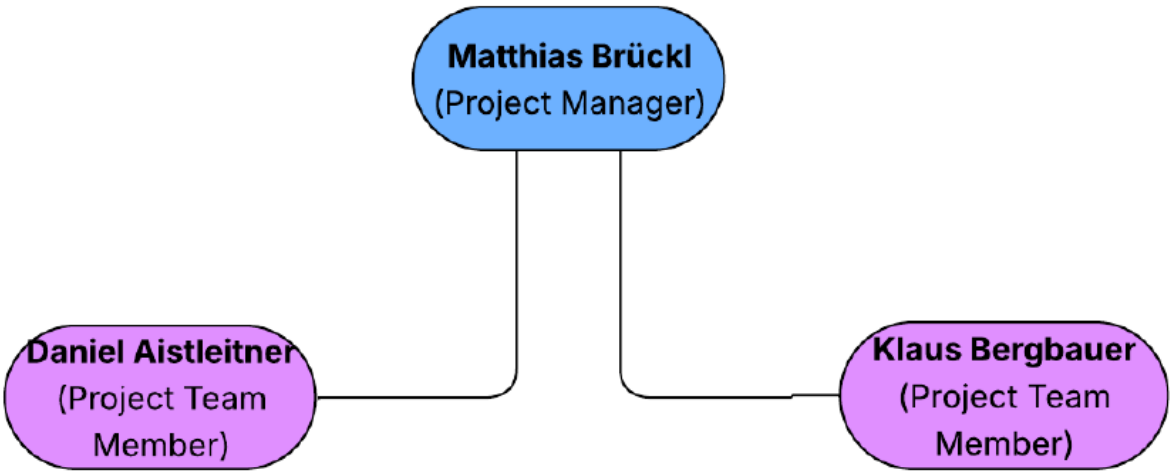
Environment	**Relationship** (potential/conflict)	**Measures**	**Who / when** **PSP Code**
Stakeholder	potential	Can help us to optimize our project	HTL-Wels/ Professors
Other Projects	potential	can help each other	Smart-Flowerpot/ Smart-Wand
Technology	potential	help us to implement it as we envision it	Database/Coding languages/ Raspberry Pi
Other Projects	conflict	Are competition	Indoor Wetterstation

1.5 Relationship to Other Projects and the Organisations´s Strategy

Wetterstation 001	RELATIONSHIP TO OTHER PROJECTS		
Programs/ Projects/	**Relationship** (potential/conflict)	**Sanctions**	**Who / when** **WBS Code**
Indoor-Wetterstation	Conflict	Competition	Paul Gnadlinger; Nicolas Goiginger
Smarter Blumentopf	Potential	Support with Know-how	Paul Auer
Smart-Wand	Potential	Support with Know-how	Thomas Toferer

Wetterstation 001	CONNECTION TO THE ORGANISATION´S STRATEGY
Strategy	**Description of connection/relationship**
-	-

1.6 Project Organisation Chart



Wetterstation 001	PROJECT-ORGANISATION	
Role in Project	**Field of duties/Skills**	**Name**
Project owner	Financial providence	HTL Wels
Project manager	Backend development	Matthias Brückl
Project team members	1. Frontend development 2. Hardware and 3d Modelling	1) Daniel Aistleitner 2) Klaus Bergbauer
Project members		

1.7 Plan of Objects of Consideration of the Project

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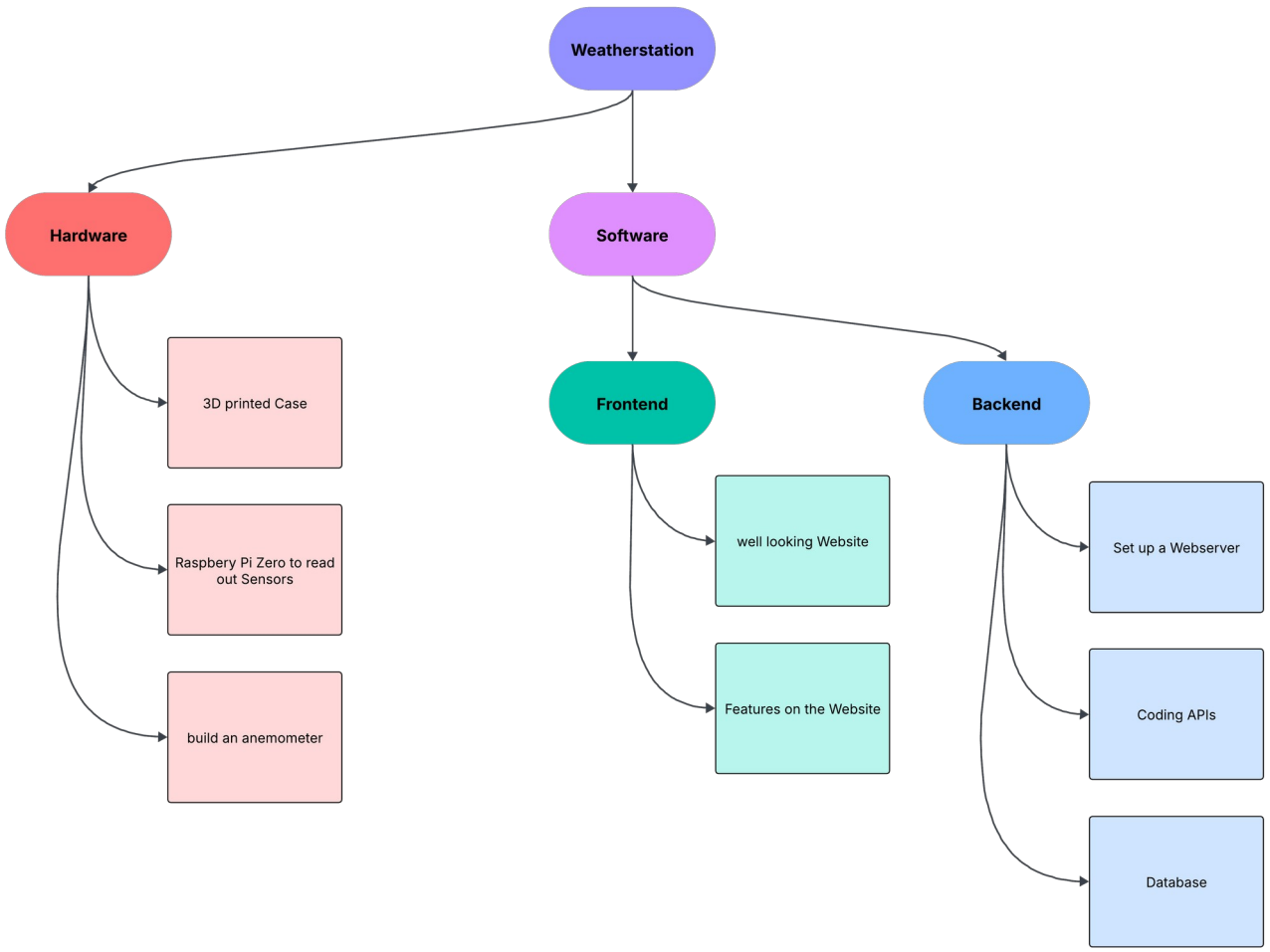
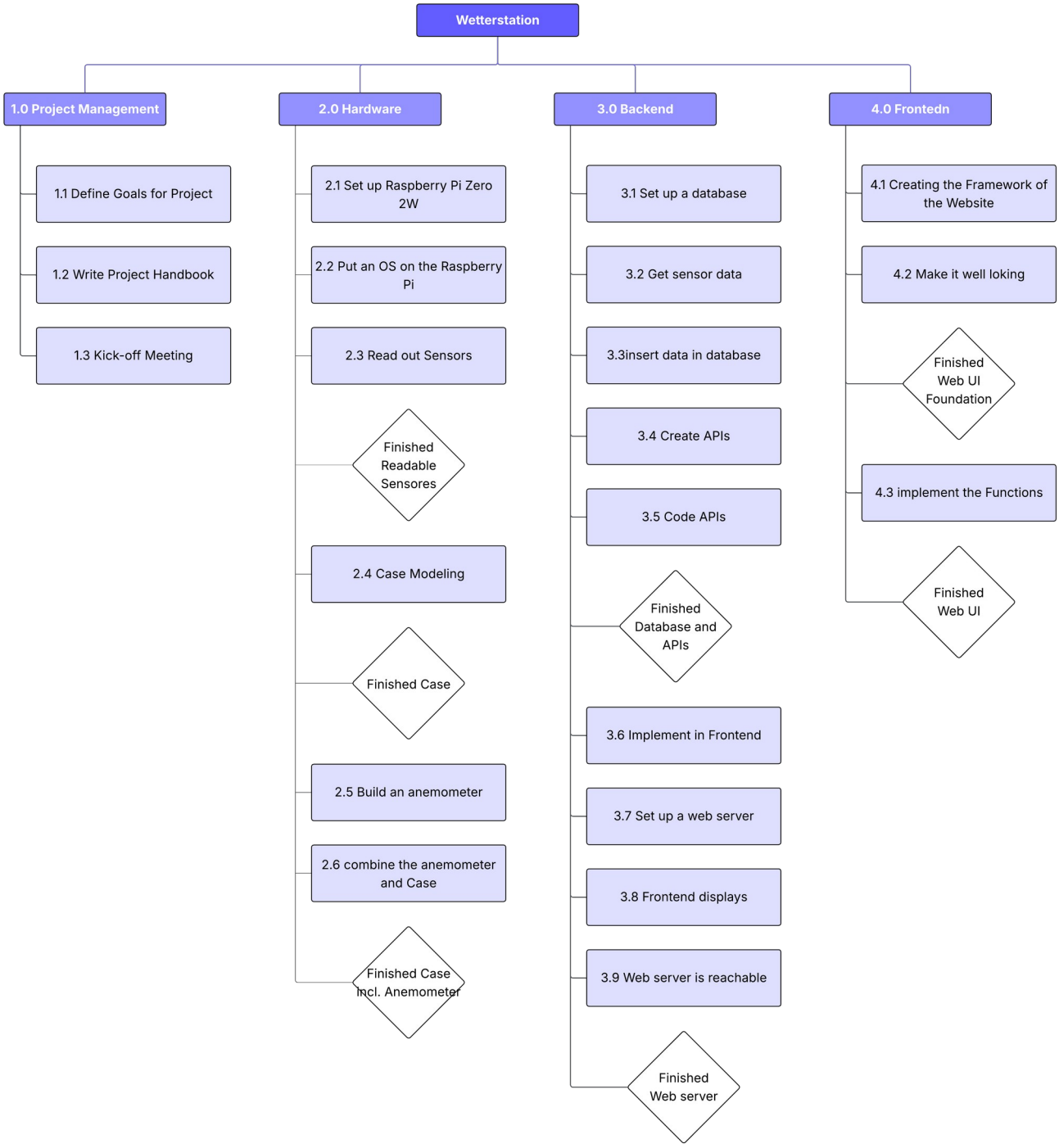
PLAN OF OBJECTS OF CONSIDERATION



1.8 Work Breakdown Structure (WBS)



1.9 Project Work-Package Specification

Wetterstation 001	PROJECT WORK-PACKAGE SPECIFICATION
1.1 Setup a Folder Structure	**WP Content** *(What shall be done?)* • Think about a clear folder structure
	WP Result *(What is achieved after WP was finished?)* • Have a clear folder structure
	Progress Measurement *(How is progress measured?)* • Every File has his right place
1.2 Workout Project Requirements	**WP Content** *(What shall be done?)* • Think about the Project Requirements
	WP Result *(What is achieved after WP was finished?)* • Have a clear vision of the Project
	Progress Measurement *(How is progress measured?)* • The Requirements willincrease
1.3 Workout Project Handbook section 1-3	**WP Content** *(What shall be done?)* • Filling the Project-Handbook
	WP Result *(What is achieved after WP was finished?)* • The sections 1-3 have the right entries
	Progress Measurement *(How is progress measured?)* • The Project-Handbook are filling
1.4 Workout Project Handbook section 4-5	**WP Content** *(What shall be done?)* • Filling the Project-Handbook section 4-5
	WP Result *(What is achieved after WP was finished?)* • The sections 4-5 have the right entries
	Progress Measurement *(How is progress measured?)* • The Project-Handbook are filling
1.5 Workout Project Specification Section	**WP Content** *(What shall be done?)* • Filling the Project Specification
	WP Result *(What is achieved after WP was finished?)* • Have a clear Project Specification, which has the right entries
	Progress Measurement *(How is progress measured?)* • The Project Specification are filling

2.1 Raspberry activation	**WP Content** *(What shall be done?)* • Setting up the raspberry
	WP Result *(What is achieved after WP was finished?)* • Make the raspberry useable
	Progress Measurement *(How is progress measured?)* • If raspberry is ready
2.2 Sensor implementation	**WP Content** *(What shall be done?)* • Sensor implementing
	WP Result *(What is achieved after WP was finished?)* • Make the raspberry useable for the sensors
	Progress Measurement *(How is progress measured?)* • If raspberry is ready
2.3 Write Sensor Scripts	**WP Content** *(What shall be done?)* • Make the data from the sensors usable
	WP Result *(What is achieved after WP was finished?)* • The data is displayed in the terminal
	Progress Measurement *(How is progress measured?)* • Scripts are finished for each sensors
2.4 Case modeling	**WP Content** *(What shall be done?)* • Modeling the case of the weather station
	WP Result *(What is achieved after WP was finished?)* • Have a file to print
	Progress Measurement *(How is progress measured?)* • Having a promising model
2.5 Print Case	**WP Content** *(What shall be done?)* • Print the Case
	WP Result *(What is achieved after WP was finished?)* • Having a ready to use case for the weather station • Combination with raspberry and other components
	Progress Measurement *(How is progress measured?)* • Having a successful print • Making the other components fit
2.6 Think about an Anemometer	**WP Content** *(What shall be done?)* • Think about an Anemometer
	WP Result *(What is achieved after WP was finished?)* • Have an idea to build an Anemometer
	Progress Measurement *(How is progress measured?)* • The concept can be implemented
2.7 Combine the	**WP Content** *(What shall be done?)*

Anemomter and the Case	• Assemble the Anemometer and the Case
	WP Result *(What is achieved after WP was finished?)* • The Raspberry Pi Zero can read the data of the anemometer
	Progress Measurement *(How is progress measured?)* • The Data is displayed in the database
3.1. Basic Framework	**WP Content** *(What shall be done?)* • Programming the Basic Framework of the Frontend
	WP Result *(What is achieved after WP was finished?)* • Creating a functional running program
	Progress Measurement *(How is progress measured?)* • If there are no errors • Program is running
3.2. Website with not yet functional features	**WP Content** *(What shall be done?)* • Programming the website with the necessary features
	WP Result *(What is achieved after WP was finished?)* • Having a not yet functional website
	Progress Measurement *(How is progress measured?)* • Control and looking at the website
3.3. Functionalities	**WP Content** *(What shall be done?)* • Programming the functionalities
	WP Result *(What is achieved after WP was finished?)* • Sensor-data and printing it out
	Progress Measurement *(How is progress measured?)* • If you can see the data on the website
3.4. Design	**WP Content** *(What shall be done?)* • Design of the website **Promotion cum laude zum Dr. med. dent. an der Semmelweis Universität Budapest**
	WP Result *(What is achieved after WP was finished?)* • Having a well looking website
	Progress Measurement *(How is progress measured?)* • Adding simple elements
4.1. Database	**WP Content** *(What shall be done?)*

	• Setup a small database to save
	WP Result *(What is achieved after WP was finished?)* • Having a well modeld Database
	Progress Measurement *(How is progress measured?)* • Values can be stored effectively
4.2. APIs	**WP Content** *(What shall be done?)* • Coding APIs
	WP Result *(What is achieved after WP was finished?)* • Frontend can reach Data
	Progress Measurement *(How is progress measured?)* • Values are displayed on the Frontend
4.3. Web Server	**WP Content** *(What shall be done?)* • Setting up a web server
	WP Result *(What is achieved after WP was finished?)* • A web server is running
	Progress Measurement *(How is progress measured?)* • The website that displays the data can be accessed

1.10 Project Responsibility Matrix

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PROJECT-RESPONSIBILITY-MATRIX

WBS-Code	WP-Title / Roles & Environment	External project owner	Project owner	Project manager	Project assistant	Project team member 1	Project team member 2	Project team member 3	Project team member 4	Project team member 5	Project member 1	Project member 2	Project member 3	Project coach
1.0	**Project Management**		**I**	**R**		**C**	**C**							
1.0.1	Define goals for Project		I	R		C	C							
1.0.2	Write Project-Handbook		I	R		C	C							
1.0.3	Kick-off meeting		I	R		C	C							
1.1	**Hardware**		**I**	**C**		**I**	**R**							
1.1.1	Set up the Raspberry Pi Zero 2W		I	I		I	R							
1.1.1.1	Put an OS on the Raspberry Pi Zero		I	I		I	R							
1.1.1.2	Use correct resistors and sensors		I	I		I	R							
1.1.1.3	Read out sensors		I	C		I	R							
1.1.2	Case modeling		I	I		I	R							
1.1.3	Build Anemometer		I	I		I	R							
1.1.3.1	Connect the Anemometer		I	I		I	R							
1.2	**Backend**		**I**	**R**		**C**	**C**							
1.2.1	Set up a database		I	R		I	C							
1.2.1.1	Get Sensor Data		I	R		I	C							
1.2.1.2	Insert in data in database		I	R		I	C							
1.2.2	Create APIs		I	R		C	I							
1.2.2.1	Code APIs		I	R		I	I							
1.2.2.2	Implement in Frontend		I	C		R	I							
1.2.3	Set up a web server		I	R		C	I							
1.2.3.1	Frontend displays		I	R		C	I							
1.2.3.2	Web server is reachable		I	R		I	I							
1.3	**Frontend**		**I**	**I**		**R**	**I**							
1.3.1	Creating the Framework of the website		I	I		R	I							
1.3.2	Make it well looking		I	I		R	I							
1.3.3	Implement the functions		I	C		R	I							

Functions
R .....Responsible
C .....Contribution
I .....has to be informed

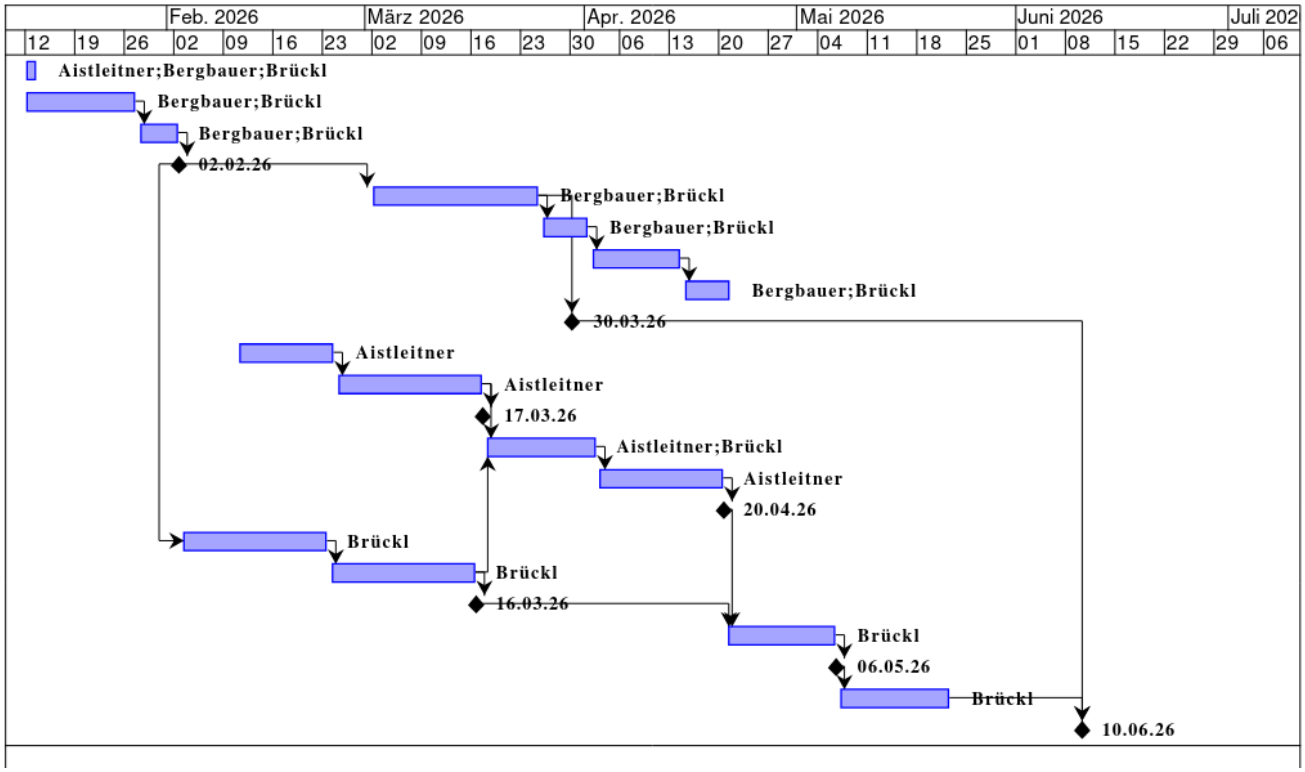
1.11 Milestoneplan

Wetterstation 001	MILESTONEPLAN			
WBS-Code	**Milestone**	**Plan date**	**Revised date**	**Actual date**
4	Finished readable sensors	12.02.2026	12.02.2026	12.02.2026
10	Finished Web UI Foundation	17.03.2026	17.03·2026	17.03·2026
7	Finished case	30.03.2026	30.3.2026	30.03.2026
16	Finished Database and APIs	26.03.2026	03.06.26	03.06.26
13	Finished Fronted	29.04.2026	03.06.26	03.06.26
18	Finished Web Server	15.05.2026	25.03.2026	25.03.2026
20	Finished Project	10.06.2026	11.06.2026	11.06.2026

*In order of plan dates.

1.12 Project Bar Chart

		Name	Dauer	Start	Ende	Vorgänger	Ressourcen
1		Software installation	2 tage?	12.01.26, 08:00	13.01.26, 17:00		Aistleitner;Bergbauer;Brückl
2		Raspberry activation	12 tage?	12.01.26, 08:00	27.01.26, 17:00		Bergbauer;Brückl
3		Sensor implementation	4 tage?	28.01.26, 08:00	02.02.26, 17:00	2	Bergbauer;Brückl
4		Finished readble sensors	0 tage?	02.02.26, 17:00	02.02.26, 17:00	3	Bergbauer;Brückl
5		Case Modeling	18 tage?	01.03.26, 08:00	25.03.26, 17:00	4	Bergbauer;Brückl
6		Think about an Anemometer	5 tage?	26.03.26, 08:00	01.04.26, 17:00	5	Bergbauer;Brückl
7		Build the Anemometer	9 tage?	02.04.26, 08:00	14.04.26, 17:00	6	
8		Combine the Anemomter and the Case	5 tage?	15.04.26, 08:00	21.04.26, 17:00	7	Bergbauer;Brückl
9		Finished Case and power supply	0 tage?	30.03.26, 07:00	30.03.26, 08:00	5	Bergbauer;Brückl
10		Framework for Web UI	10 tage?	11.02.26, 08:00	24.02.26, 17:00		Aistleitner
11		Web UI with not yet functional features	15 tage?	25.02.26, 08:00	17.03.26, 17:00	10	Aistleitner
12		Finished Web UI Foundations	0 tage	17.03.26, 17:00	17.03.26, 17:00	11	Aistleitner
13		Functionalities in Web UI	12 tage?	18.03.26, 08:00	02.04.26, 17:00	11;17	Aistleitner;Brückl
14		Design	12 tage?	03.04.26, 08:00	20.04.26, 17:00	13	Aistleitner
15		Finished Frontend	0 tage	20.04.26, 17:00	20.04.26, 17:00	14	Aistleitner;Brückl
16		Setup a Database	15 tage?	03.02.26, 08:00	23.02.26, 17:00	4AA	Brückl
17		Creating APIs	15 tage?	24.02.26, 08:00	16.03.26, 17:00	16	Brückl
18		Finished Database&APIs	0 tage	16.03.26, 17:00	16.03.26, 17:00	17	Brückl
19		Create Web Server	12 tage?	21.04.26, 08:00	06.05.26, 17:00	15;18	Brückl
20		Finished Web Server	0 tage	06.05.26, 17:00	06.05.26, 17:00	19	Brückl
21		setup mDNS	12 tage	07.05.26, 08:00	22.05.26, 17:00	20	Brückl
22		Finished Project	0 tage	10.06.26, 07:00	10.06.26, 08:00	9;21	Aistleitner;Bergbauer;Brückl



1.13 Resource Plan

Wetterstation 001	Resource Plan					
WBS-Code	**Phase/Work-package**	**Type of resource**	**Planned quantity (days)**	**Revised quantity (days)**	**Actual quantity (days)**	**Deviation (days)**
2	Raspberry activation	Raspberry Pi Zero 2W, Laptop, Raspberry Pi Imager	12	12	12	0
3	Sensor implementation	Raspberry Pi Zero 2W, DHT22, BME280	12	12	12	0
6	Think about an Anemometer	Raspberry Pi Zero 2W, Parts of an Anemometer	5	5	5	0
7	Build an Anemometer	Parts of an Anemometer	10	10	10	0
5	Case modeling and printing	3D-Printer, ABS-Filament, Laptop, Oneshape	15	15	15	0
8	Combine Anemometer and Case	Case, Anemometer	10	10	10	0
10	Framework for Web UI	Laptop, Virtual Studio Code	10	10	10	0
11	Web UI with not yet functional features	Laptop, Virtual Studio Code	15	15	15	0
13	Functionalities in Web UI	Laptop, Virtual Studio Code	12	12	12	0
14	Web UI Design	Laptop, Virtual Studio Code	12	12	12	0
16	Setup a Database	Laptop, Raspberry Pi Zero 2W, DataGrip	15	15	15	0
17	Creating APIs	Laptop, Raspberry Pi Zero 2W, Virtual Studio Code	15	15	15	0
19	Create Web Server	Laptop, Raspberry Pi Zero 2W, Virtual Studio Code	12	12	12	0
21	Setup mDNS	Laptop, Raspberry Pi Zero 2W, Virtual Studio Code	12	12	12	0

1.14 Project Cost Plan

Wetterstation 001	PROJECT COST PLAN				
WBS-Code, WP-Title	Type of Cost	Planned cost	Revised cost	Actual cost	Deviation
3	• Personnel	0€			
	• Material	98€			
	• External services	0€			
	• Other	0€			
	Total	98€			
	•				
	•				
	•				
	•				
	•				
	•				
	•				
	•				
Project cost		98€			

1.15 Project Communication

Wetterstation 001	PROJECT-COMMUNICATION			
Title	**Objectives, Content**	**Participants**	**Schedule**	**Location**
Project owner meeting	• project status • decisions • acceptance of progress report	Project owner, Project manager	Not yet scheduled	Teacher Table
Project controlling meeting	• project status • controlling of tasks, schedule, resources, costs • controlling of project environments • social controlling • prepare proposal for decision	Project manager, Project team, Project coach	Every Monday at 11 o clock	Last Row, classroom T233
Subteam meeting	• Coordination of subteams • Discussion of problems	Subteam	Every Monday at 11 o clock	Last Row, classroom T233

1.16 Project „Rules“

Communication:

- **Weekly Meetings:** The participation of the Project Team Member and the Project Manager in the Weekly Meetings is very important. The meeting is scheduled every Monday at 11 o clock.
- **File sharing:** All files belonging to the project must be published in the GitHub repository. No other ways to share files are allowed.
- **Communication channels:** All important information will be published in the team group.

General Collaboration:

- **Time Management:** Each team member is responsible for their own work packages. These must be completed by the deadline. If more time is needed, this must be requested promptly.
- **Support:**
  - If help is needed with a work package, it must be requested promptly.
  - If someone needs help and you have the time and are able to help, you are obligated to do so.

Documentation:

- **Version Control:** The Project Handbook must be updated regularly. Significant changes require a new version number and an entry in the document history.

Technical Standard:

- **Hardware Care:** Care must be taken with the hardware components, and they must not be damaged or broken in any way. If this happens, the affected part must be replaced.

1.17 Project Risk Analysis

		PROJECT RISK ANALYSIS							
WBS-Code	WBS-description	Risk-description	Priority	Benefit /Cost	Probability	Risk-value	Delay	Avoiding action	Cost for minimizing risk
(Code)	*(Text)*	*(Text)*	*(Range)*	*(iuro)*	*(Percentage)*	*(iuro)*	*(iWeeks)*	*(Text)*	*(Euro)*
2	Raspberry activation	break	1	30€	10%	30€	1-2Weeks	Take Care	0€
3	Senor implementation	Sensor breaks	1	5-10€	10%	5-10€	1-2Weeks	Take Care	0€
5	Workout energy independent supply	Provide not enough electricity	2	20€	20%	20€	1-2Weeks	First Think, than buy	15€
6	Case Modeling	Wrong size	4	5€	40%	5€	½ Week	Measure correctly	0€
14	Setup a Database	Database is too small	2	0€	10%	0€	½ Week	Think first about the right size	0€
15	Creating APIs	APIs don't share the data	5	0€	15%	0€	½ Week	Have a clear structure	0€
17	Create Web Server	The server don't start	3	0€	10%	0€	½ Week	Inform you how to create a server	0€
18	setup mDNS	DNS doesn't work	3	0€	20%	0€	½ Week	Inform you how to setup mDNS	0€
Project total									

1.18 Project Documentation

Area	Description
File	In the GitHub Repository: Directory named “Documentation”
Access Authorisation	Everyone from the Project Team has access to the documentation
Naming convention	DDMMYYYY_version_name.docx
Rules	

2 Project Start

2.1 Minutes– Project Start

2.1.1 Project Start Workshop

2.1.2 Follow-up Workshop

2.1.3 Project Owner Meeting

3 Project Co-ordination

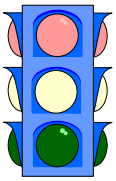
3.1 Approval of Work-packages

Wetterstation 001	APPROVAL OF WORK-PACKAGES				
WBS-Code	**Work-package**	**WP-Owner**	**Date**	**Approval by**	**Signature**
2	Raspberry activation	Berbauer K.	27.01.26	Brückl M.	Bρ⌉χκλ M.
3	Sensor implementation	Berbauer K.	02.02.26	Brückl M.	Bρ⌉χκλ M.
5	Case Modeling	Berbauer K.	19.04.26	Brückl M.	Bρ⌉χκλ M.
6	Think about an Anemometer	Berbauer K.	25.03.26	Brückl M.	Bρ⌉χκλ M.
7	Build the Anemometer	Bergbauer K.	20.05.26	Brückl M.	Bρ⌉χκλ M.
10	Framework for Web UI	Aistleitner D.	10.03.26	Brückl M.	Bρ⌉χκλ M.
11	Web UI with not yet functional features	Aistleitner D.	17.03.26	Brückl M.	Bρ⌉χκλ M.
13	Functionalities in Web UI	Aistleitner D.	27.05.26	Brückl M.	Bρ⌉χκλ M.
14	Design	Aistleitner D.	25.04.26	Brückl M.	Bρ⌉χκλ M
16	Setup a Database	Brückl M.	15.03.26	Brückl M.	Bρ⌉χκλ M.
17	Creating APIs	Brückl M.	20.05.26	Brückl M.	Bρ⌉χκλ M.
19	Create Web Server	Brückl M.	11.03.36	Brückl M.	Bρ⌉χκλ M.
21	setup mDNS	Brückl M.	30.05.26	Brückl M.	Bρ⌉χκλ M.

3.2 Minutes – Project Co-ordination

4 Project Controlling

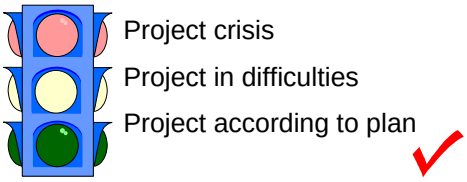
4.1 Project Status Report

Wetterstation 001	PROJECT STATUS REPORT as of Bergbauer Klaus
 Project crisis Project in difficulties Project according to plan ✓	• **Overall Status:** • The project is progressing well. Raspberry Pi Zero 2W has been successfully set up with SSH and Wi-Fi enabled. DHT11 sensor testing is underway, and the basic React website structure is complete.

2) Status Project objectives • Set up Raspberry Pi Zero 2W with OS Lite • Enable remote access via SSH • Test DHT11 temperature/humidity sensor • Develop React-based web interface	**Activities:** • Raspberry Pi successfully configured • DHT11 sensor integrated and tested (command line output working) • Basic website structure implemented in React
3) Status Project progress • Hardware setup: 80% complete (waiting for hardware shipment) • Sensor integration: 55% complete (testing phase) • Website development: 20% complete (structure done, output pending)	**Activities:** • Raspberry OS Lite installed on 32GB Micro SD card • SSH and Wi-Fi connectivity established • DHT11 sensor displaying data in terminal • React frontend Framework implemented
4) Status Schedule • On track - all milestones met by 03.02.2026 • No delays reported	**Activities:** • Phase 1 (Hardware Setup) completed on time (with lend Raspberry) • WBS 3 (Sensor Testing) in progress as planned
5) Status Resources/costs • Raspberry Pi Zero 2W (lended) • 32GB Micro SD card • DHT11 sensor (for testing) • DHT22 sensor • BMP280	**Activities:** • Nearly all required hardware procured • Budget within limits • No additional resource requirements identified
6) Status Context • Raspberry Pi Imager used for OS installation • PowerShell used for remote SSH connection	**Activities:** • Documentation of setup procedure completed • Network connectivity stable • No external blockers or dependencies
7) Status Organisation/culture • Author: Klaus Bergbauer • Collaboration ongoing	**Activities:** • Knowledge sharing via progress documentation • Testing and development workflow established • Communication channels functional (SSH, Wi-Fi)

4.2 Project Status Report

Wetterstation 001	PROJECT STATUS REPORT as of Brückl Matthias

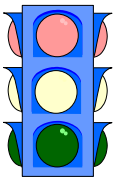


1) Overall Status:

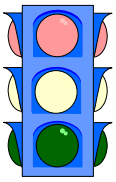
- The project is progressing well. Raspberry Pi Zero 2W can readout the DHT22 and the BME280 sensors. The first steps towards a database have been taken and the Web User Interface has been optimized.

2) Status Project objectives • Integrate high-precision sensors (DHT22, BME280) • Automate data retrieval via scripting • Improve React frontend design	**Activities:** • DHT22 & BME280 successfully wired and recognized • Custom script for data reading completed • UI styling and layout optimization finished
3) Status Project progress • Hardware setup: 100% (All sensors arrived and installed) • Data Processing: 90% (Scripts functional) • Website development: 40% (UI refined)	**Activities:** • Replaced DHT11 with superior DHT22/BME280 sensors • Developed and debugged data collection script • Implemented CSS/styling updates to the React WebUI
4) Status Schedule • On track - all milestones met by 16.02.2026 • No delays reported	**Activities:** • WP 2 (Raspberry activation) completed on time (with lend Raspberry) • WP 3 (Sensor implementation) completed
5) Status Resources/costs • Raspberry Pi Zero 2W (lended) • 32GB Micro SD card • DHT22 sensor • BME280	**Activities:** • Nearly all required hardware procured • Budget within limits • No additional resource requirements identified
6) Status Context • PowerShell used for remote SSH connection • Python Scripts used to read out the sensors • data output verified via terminal	**Activities:** • Documented the pinout configuration for the new sensors • No external blockers or dependencies
7) Status Organisation/culture • Author: Matthias Brückl • Collaboration ongoing • Documentation updated for new hardware	**Activities:** • Knowledge sharing via progress documentation • Testing and development workflow established • new Scripts are published to the other Projektmembers

4.3 Project Status Report

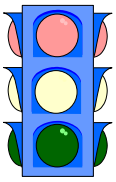
Wetterstation 001	PROJECT STATUS REPORT as of Aistleitner Daniel
 Project crisis Project in difficulties Project according to plan ✓	1) **Overall Status:** • The project is progressing well. Work on the system architecture has expanded. We have successfully started the database initialization and the independent power concept, necessitating an earlier setup of the web server for performance monitoring.
2) Status Project objectives • Worked out a concept for an independent power supply • Write a logging script • created a web server on the Raspberry Pi	**Activities:** • Concept design for solar/battery power integration • Development of a logging strategy for system stability • Initial web server deployment
3) Status Project progress • WP5 (Work out energy independent power supply) completed • WP14 (Setup a Database) is 95% finished • WP9 (Web UI with not yet functional features) is 75% finished	**Activities:** • Research and component selection for the independent power supply • An Apache Web Server was installed on the Raspberry Pi • Setup an SQLite Database
4) Status Schedule • On track • minimal delays reported • The web server setup was pulled forward from WP 17 to analyze system load early	**Activities:** • all milestones are almost reached by 11.03.2026
5) Status Resources/costs • Raspberry Pi Zero 2W (lended) • 32GB Micro SD card • DHT22 sensor • BME280	**Activities:** • Budget within limits • Solar panel and batteries are being ordered •
6) Status Context • Used Python for the logging script and SQL to build a database • Used the Apache services for the web server	**Activities:** • The scripts were documented with comments • documented all commands for the installation of the web services
7) Status Organisation/culture • Author: Aistleitner Daniel • Collaboration ongoing • Documentation updated	**Activities:** • Knowledge sharing via progress documentation •

4.4 Project Status Report

Wetterstation 001	PROJECT STATUS REPORT as of Brückl M.
 Project crisis Project in difficulties Project according to plan ✓	1) **Overall Status:** • The project is progressing well. The backend foundation has been finalized: a main.py entry point and a systemd daemon on the Raspberry Pi have been set up to automatically start the backend. The database initialization script is now complete. The frontend has been redesigned with a dashboard-style layout. A hardware change was made: solar power has been dropped in favor of an anemometer for wind speed measurement. Power supply concepts using a battery have been considered.

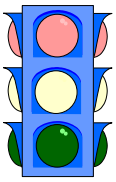
2) Status Project objectives • Set up main.py as system entry point and configure systemd daemon • Finalize database initialization script (init_db.py) • Redesign frontend with a dashboard-style layout • Replace solar power with anemometer for wind speed measurement • Evaluate battery-based power supply concept	**Activities:** • main.py created with logging and DB initialization on startup • Systemd daemon configured to auto-start backend on Raspberry Pi boot • init_db.py finalized: tables, indices, and seed data completed • Frontend redesigned as a dashboard with improved data visualization
3) Status Project progress • WP14 (Setup a Database): 100% complete • WP9 (Web UI with not yet functional features): 100% complete • WP5 (Energy independent power supply): concept revised, solar dropped	**Activities:** • Database init script fully implemented and tested • New dashboard layout deployed on frontend • Hardware concept updated: anemometer added, solar panel removed
4) Status Schedule • On track. No significant delays.	**Activities:** • WP8 (Framework for Web UI) completed • WP9 (Web UI with not yet functional features) completed • WP14 (Setup a Database) completed
5) Status Resources/costs • Budget within limits. Anemometer to be ordered.	**Activities:** • Budget within limits • Solar panel dropped from requirements • Anemometer to be ordered
6) Status Context • Python (backend) • SQLite • React + TypeScript (frontend)	**Activities:** • Backend scripts documented with comments • Dashboard layout implemented in React • Systemd service file created for auto-start
7) Status Organisation/culture • Author: Matthias Brückl. • Collaboration ongoing. • Documentation updated.	**Activities:** • Knowledge sharing via progress documentation • New scripts published to project members via GitHub

4.5 Project Status Report

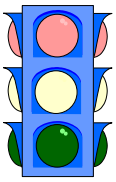
Wetterstation 001	PROJECT STATUS REPORT as of Bergbauer K.
 Project crisis Project in difficulties Project according to plan ✓	1) **Overall Status:** • The project is progressing well. Parts for the self-built anemometer have been researched and ordered. The folder structure on the Raspberry Pi has been reorganized for better maintainability. The frontend development has been continued and the Project Handbook has been updated to reflect the hardware change from solar power to an anemometer.

2) Status Project objectives • Research and order parts for a self-built anemometer • Reorganize folder structure on Raspberry Pi • Continue frontend development	**Activities:** • Identified required components for the anemometer and placed the order • Revised and cleaned up the directory structure on the Raspberry Pi • Continued work on the website frontend
3) Status Project progress • WP6 (Think about an Anemometer): 100% complete • WP7 (Build the Anemometer): 10% (parts ordered, awaiting delivery) • WP13 (Functionalities in Web UI): in progress • WP14 (Web UI Design): in progress	**Activities:** • Anemometer parts sourced and ordered • Raspberry Pi file system restructured for cleaner project layout • Frontend development ongoing
4) Status Schedule • On track. No significant delays.	**Activities:** • WP6 (Think about an Anemometer) completed • on time Parts for WP7 ordered as planned
5) Status Resources/costs • Budget within limits. Anemometer is ordered	**Activities:** • Budget within limits • Anemometer components ordered • Solar panel removed from requirements
6) Status Context • React + TypeScript (frontend) • Raspberry Pi Zero 2W	**Activities:** • Frontend continued to be developed on local machine
7) Status Organisation/culture • Author: Bergbauer Klaus • Collaboration ongoing. • Documentation updated.	**Activities:** • Knowledge sharing via progress documentation •

4.6 Project Status Report

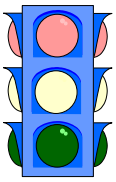
Wetterstation 001	PROJECT STATUS REPORT as of Aistleitner D.
 Project crisis Project in difficulties Project according to plan ✓	• **Overall Status:** • The project is progressing steadily. This week was focused on hardware preparation. The ordered anemometer parts have arrived and been received. The components intended for 3D printing have been commissioned. Additionally, work on the 3D modeling of the housing has begun using Tinkercad.
2) Status Project objectives • Receive and inspect ordered anemometer parts • Commission 3D-printable components • Begin 3D modeling of the weather station case	**Activities:** • Anemometer parts received and inspected Components to be 3D-printed have been commissioned • Started 3D design of the case in Tinkercad
3) Status Project progress • WP7 (Build the Anemometer): 20% (parts received, assembly pending) • WP5 (Case Modeling): 15% (3D model started in Tinkercad) • WP8 (Print Case): 5% (print job commissioned)	**Activities:** • Anemometer components checked for completeness 3D printing order placed for required parts Initial case geometry drafted in Tinkercad
4) Status Schedule • On track. No significant delays.	**Activities:** • Parts arrived as expected 3D modeling started on schedule
5) Status Resources/costs • Budget within limits.	**Activities:** • Budget within limits All required anemometer parts received 3D print material (ABS) in use
6) Status Context • Tinkercad used for 3D modeling Anemometer parts on hand	**Activities:** • 3D model being developed locally in Tinkercad • Components organized and stored properly
7) Status Organisation/culture • Author: Aistleitner Daniel • Collaboration ongoing. • Documentation updated.	**Activities:** • Knowledge sharing via progress documentation Updates shared with project team via GitHub

4.7 Project Status Report

Wetterstation 001	PROJECT STATUS REPORT as of Brückl M.
 Project crisis Project in difficulties Project according to plan ✓	• **Overall Status:** • The project is progressing well. The 3D-printed components for the anemometer have arrived and assembly can now begin. Work on the case design has been started. API development has been initiated to connect the backend with the frontend. The web interface has been further developed.

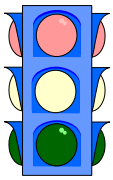
2) Status Project objectives • Receive 3D-printed anemometer components and prepare assembly • Start drawing/designing the weather station case • Begin API development • Continue web interface development	**Activities:** • 3D-printed parts for the anemometer received and ready for assembly • Case design work started • API development initiated • Web UI further developed and improved •
3) Status Project progress • WP7 (Build the Anemometer): 60% (3D parts received, assembly pending) • WP5 (Case Modeling): 30% (design started) • WP17 (Creating APIs): 20% (development initiated) • WP13/14 (Web UI Functionalities & Design): in progress	**Activities:** • Anemometer assembly preparation underway • Case geometry being designed • First API endpoints being coded • Frontend continued to be developed
4) Status Schedule • On track. No significant delays.	**Activities:** • All work packages progressing as planned
5) Status Resources/costs • Budget within limits.	**Activities:** • No additional costs incurred
6) Status Context • React + TypeScript (frontend) • Python (backend/APIs) • Tinkercad (case design)	**Activities:** • Development ongoing on local machines and loan Raspberry Pi Zero 2W
7) Status Organisation/culture • Author: Brückl M. • Collaboration ongoing. • Documentation updated.	**Activities:** • Updates shared with project team via GitHub

4.8 Project Status Report

Wetterstation 001	PROJECT STATUS REPORT as of Bergbauer K.
 Project crisis Project in difficulties Project according to plan ✓	• **Overall Status:** • The project is progressing well and according to plan. A major technical milestone was reached by configuring the Raspberry Pi to create its own hotspot for WLAN configuration when no network is available. The hardware phase has moved into physical testing with the first test prints of the case and the start of the anemometer assembly. Software development is also on track with the addition of a settings window to the web interface and continued API development. The team successfully received the Raspberry Pi Zero 2W.

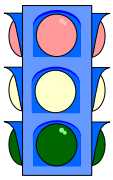
2) Status Project objectives • Implement a hotspot mode on the Raspberry Pi for easy WLAN setup. • Continue case modeling and evaluate initial test prints. • Expand the web interface with a dedicated settings window. • Advance the development of backend APIs. • Start the physical assembly of the anemometer.	**Activities:** • Raspberry Pi hotspot mode configured and tested • First test prints of the case completed and evaluated • Settings window added to the web interface • API development continued and expanded • Raspberry Pi Zero 2W received by the team • Anemometer assembly started
3) Status Project progress • WP7 (Build the Anemometer): 70% (assembly started) • WP5 (Case Modeling): 45% (first test prints completed) • WP17 (Creating APIs): 40% (development ongoing) • WP13/14 (Web UI Functionalities & Design): in progress (settings window added)	**Activities:** • Hotspot functionality implemented on Raspberry Pi • Initial case test prints reviewed and assessed • Settings window integrated into the web interface • Additional API endpoints developed • Own Raspberry Pi Zero 2W received and set up • Anemometer physical assembly begun
4) Status Schedule • On track. No significant delays.	**Activities:** • All work packages progressing as planned
5) Status Resources/costs • Budget within limits.	**Activities:** • No additional costs incurred
6) Status Context • React + TypeScript (frontend / web interface) • Python (backend / APIs) • Tinkercad / CAD tool (case modeling) • Raspberry Pi Zero 2W (own hardware)	**Activities:** • Hotspot configuration documented • Test print results evaluated and recorded • Development ongoing on local machines and own Raspberry Pi Zero 2W
7) Status Organisation/culture • Author: Bergbauer K. • Collaboration ongoing. • Documentation updated.	**Activities:** • Updates shared with project team via GitHub

4.9 Project Status Report

Wetterstation 001	PROJECT STATUS REPORT as of Aistleitner D.
 Project crisis Project in difficulties Project according to plan ✓	1) **Overall Status:** • The project is progressing well and according to plan. The APIs have been completed and successfully integrated into the frontend. The web interface has been visually improved and now displays more sensor data. First steps towards a WLAN configuration feature in the frontend have been taken. Additionally, the first sensors are now delivering live data into the database.

2) Status Project objectives • Complete and integrate APIs into the frontend • Improve web interface design and extend data display • Begin WLAN configuration feature in frontend • Connect first sensors to the database pipeline	**Activities:** • APIs fully developed and connected to the frontend. • Frontend redesigned and extended to show more sensor values. • Initial implementation of a WLAN settings feature started. • First sensor readings successfully written to the database.
3) Status Project progress • WP17 (Creating APIs): 100% complete • WP13 (Functionalities in Web UI): 80% • WP14 (Web UI Design): 70% • WP21 (Setup mDNS / WLAN): 15% (first steps taken) • WP16 (Database): 90% (live sensor data incoming)	**Activities:** • APIs developed, tested and connected to frontend • Dashboard updated to show all available sensor readings • Design improvements applied across the web interface • Database successfully receiving live sensor data • WLAN configuration feature under early development
4) Status Schedule • On track. No significant delays.	**Activities:** • All work packages progressing as planned • API milestone reached on schedule
5) Status Resources/costs • Budget within limits.	**Activities:** • No additional costs incurred
6) Status Context • React + TypeScript (frontend) • Python (backend/APIs) • SQLite (database) • Raspberry Pi Zero 2W	**Activities:** • Development ongoing on local machines and own Raspberry Pi Zero 2W API and frontend integration documented
7) Status Organisation/culture • Author: Aistleitner D. • Collaboration ongoing • Documentation updated	**Activities:** • Updates shared with project team via GitHub

4.10 Project Status Report

Wetterstation 001	PROJECT STATUS REPORT as of Brückl M.
 Project crisis Project in difficulties Project according to plan ✓	1) **Overall Status:** • The project is nearing completion. The web interface has been finalized, the backend is fully implemented, and the case design is almost done with only minor adjustments remaining.

2) Status Project objectives • Finalize the web interface • Complete the backend implementation • Finalize the case with minor adjustments	**Activities:** • Web interface fully developed and functional • Backend completed and running stable on the Raspberry Pi • Case nearly finished, small modifications still pending
3) Status Project progress • WP13 (Functionalities in Web UI): 100% complete • WP14 (Web UI Design): 100% complete • WP17 (Creating APIs): 100% complete • WP5 (Case Modeling): 90% (minor adjustments pending) • WP2 (Print Case): 90% (final print pending)	**Activities:** • All frontend features implemented and tested • Backend fully integrated with database and sensors • Case model reviewed, small design corrections identified
4) Status Schedule • On track. Project close down on 10.06.2026 remains achievable.	**Activities:** • All major work packages completed on time
5) Status Resources/costs • Budget within limits.	**Activities:** • No additional costs incurred
6) Status Context • React + TypeScript (frontend) • Python (backend/APIs) • SQLite (database) • Raspberry Pi Zero 2W	**Activities:** • Development finalized on local machines and Raspberry Pi Zero 2W
7) Status Organisation/culture • Author: Brückl M. • Collaboration ongoing • Documentation updated	**Activities:** • Updates shared with project team via GitHub

4.11 Additional Project Status Reports

4.12 Minutes – Project controlling

4.12.1 Project Controlling Meetings

4.12.2 Project Owner Meetings

5 Project Close Down

5.1 Project Close Down Report

Wetterstation 001

PROJECT CLOSE DOWN REPORT

- **Overall impression**

The project was completed successfully. A fully functional outdoor weather station was built and deployed, including a self-built anemometer, a React-based web interface with persistent data storage, and auto-start via systemd. The final product goes beyond the original scope in some areas while other planned features were dropped or replaced.

1) **Reflection: Fulfilment of objectives**

Core objectives were met: sensors are read out and stored, the web interface visualises current and historical data, and the case is printed and assembled. The solar power supply was replaced by a battery-based concept. Wind speed was added via a custom anemometer. Air pressure (BME280) was dropped from the final build.

- **Reflection: Deliverables / Schedule**

Database/API and frontend milestones were delayed to 03.06.2026. The web server was finished ahead of schedule. Project close down on 10.06.2026 was achieved.

- **Reflection: Resources / Costs**

Total costs stayed within the planned ~98€. No unplanned additional costs occurred.

- **Reflection: Internal Organisation / Environmental Relationships**

Weekly meetings and GitHub-based file sharing worked well throughout the project. Stakeholder projects provided useful input.

- **Performance appraisal (Project owner, Project manager, Project member)**

All three team members contributed actively throughout the project. Matthias Brückl managed the backend and project coordination with additional frontend upgrading. Daniel Aistleitner led the frontend development. Klaus Bergbauer owned hardware, 3D modelling, and the anemometer build. Collaboration was effective and responsibilities were clear.

- **Lessons learned (Summary of Experiences and suggestions for improvement)**

- Define API contracts between frontend and backend early
- Clarify power supply concept at project start
- Coordinate case design with software team from the beginning
- Separate project manager and developer roles more clearly

- **Post-Project Phase Planning, Additional Tasks**

To-Do	Owner	Schedule
Present at HTL-Wels	All	22. or 24.6.26
Final documentation check	Brückl M.	
Deploy final image to GitHub	Brückl M.	
Hand over device to teacher	All	

- **Project Close Down**

HTL Wels (Project owner) *Brückl Matthias* (Project manager)

5.2 Minutes – Project Close Down

5.2.1 Project Close Down Workshop

The Project Close Down Workshop will be in Bibione, Italy!